aubergine.

Designing Reliable Connected-Device Apps with Swift & BLE

14 MAR 2026

Mobile apps are the interface to physical devices

Bluetooth devices are everywhere:

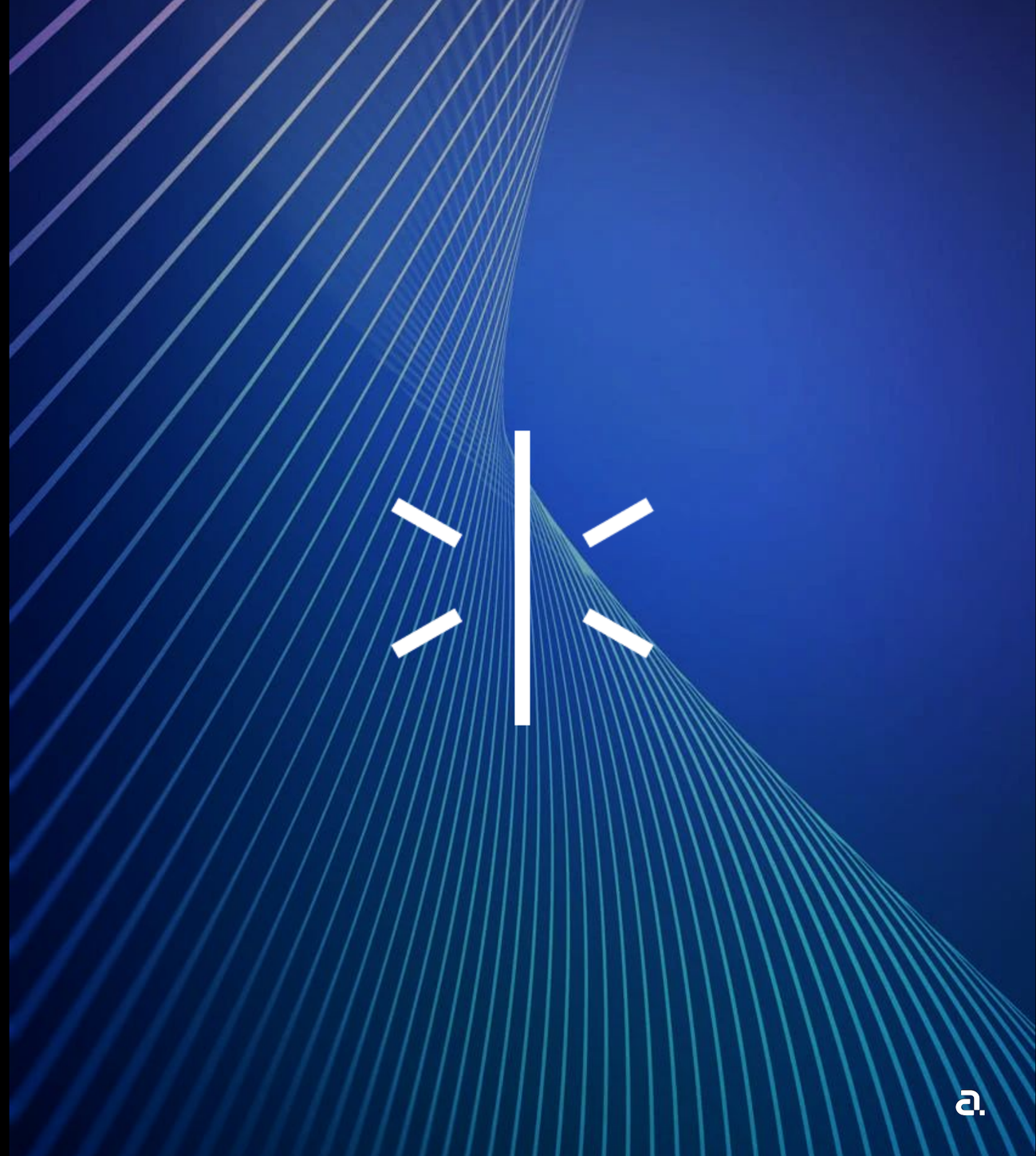
- Wearables and health devices
- Smart home accessories
- Fitness trackers

WHY BLE APPS ARE CHALLENGING

BLE demos are easy. Production BLE apps are not.

Challenges:

- Unstable wireless connections
- Limited bandwidth
- iOS background restrictions
- Firmware inconsistencies
- Battery constraints



WHAT WE'LL COVER

We'll move from BLE fundamentals to building production-ready BLE apps.

Content:

- BLE fundamentals
- BLE development in Swift
- Reliable BLE architecture
- Real-world reliability challenges
- iOS background modes & permissions
- Testing and debugging tools
- Live App Demo
- Production Lessons


01
BLE
FUNDAMENTALS

BLE trades bandwidth for extremely low power consumption.

Bluetooth Low Energy (BLE) is designed for:

- Low power consumption
- Small data packets
- Intermittent communication
- Battery-powered devices

BLE ROLES

Most mobile apps act as centrals connecting to device peripherals.

BLE communication involves two roles:

- Central → Phone
- Peripheral → Device

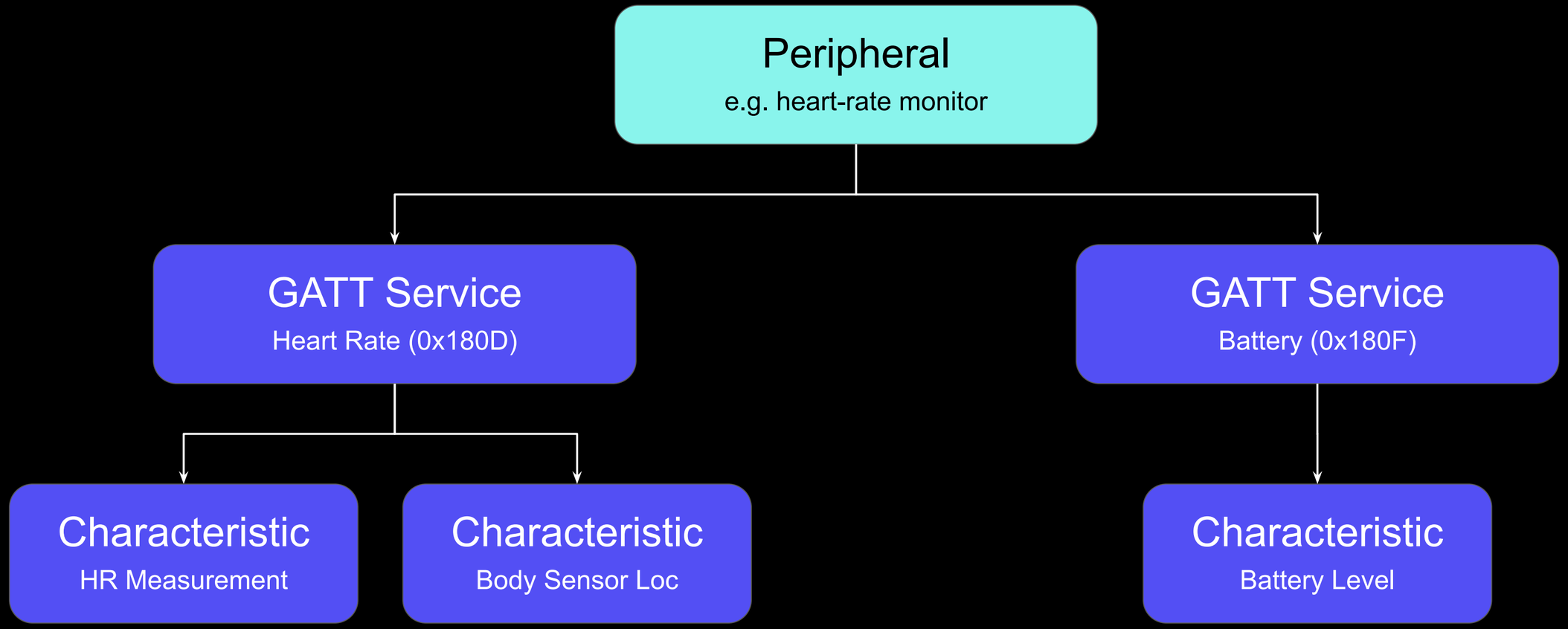
BLE DATA MODEL (GATT)

BLE devices expose functionality through services and characteristics.

BLE data is organized using GATT.

- Services
- Characteristics
- Descriptor

BLE DATA MODEL (GATT)



Properties: Read Write Notify Indicate

BLE COMMUNICATION LIFECYCLE

BLE communication follows a strict lifecycle.

Typical BLE interaction:

- Scan for nearby devices
- Discover peripheral
- Connect to device
- Discover services
- Discover characteristics
- Read / Write data
- Subscribe for notifications


02
BLE DEVELOPMENT
IN SWIFT

CoreBluetooth provides the foundation for BLE communication on iOS.

Key classes:

CBCentralManager
Manages scanning and connections

CBPeripheral
Represents the device

CBService
Represents device services

CBCharacteristic
Represents device data points

BEGINNER MISTAKE

Mixing UI and BLE logic leads to fragile apps.

Common mistake:

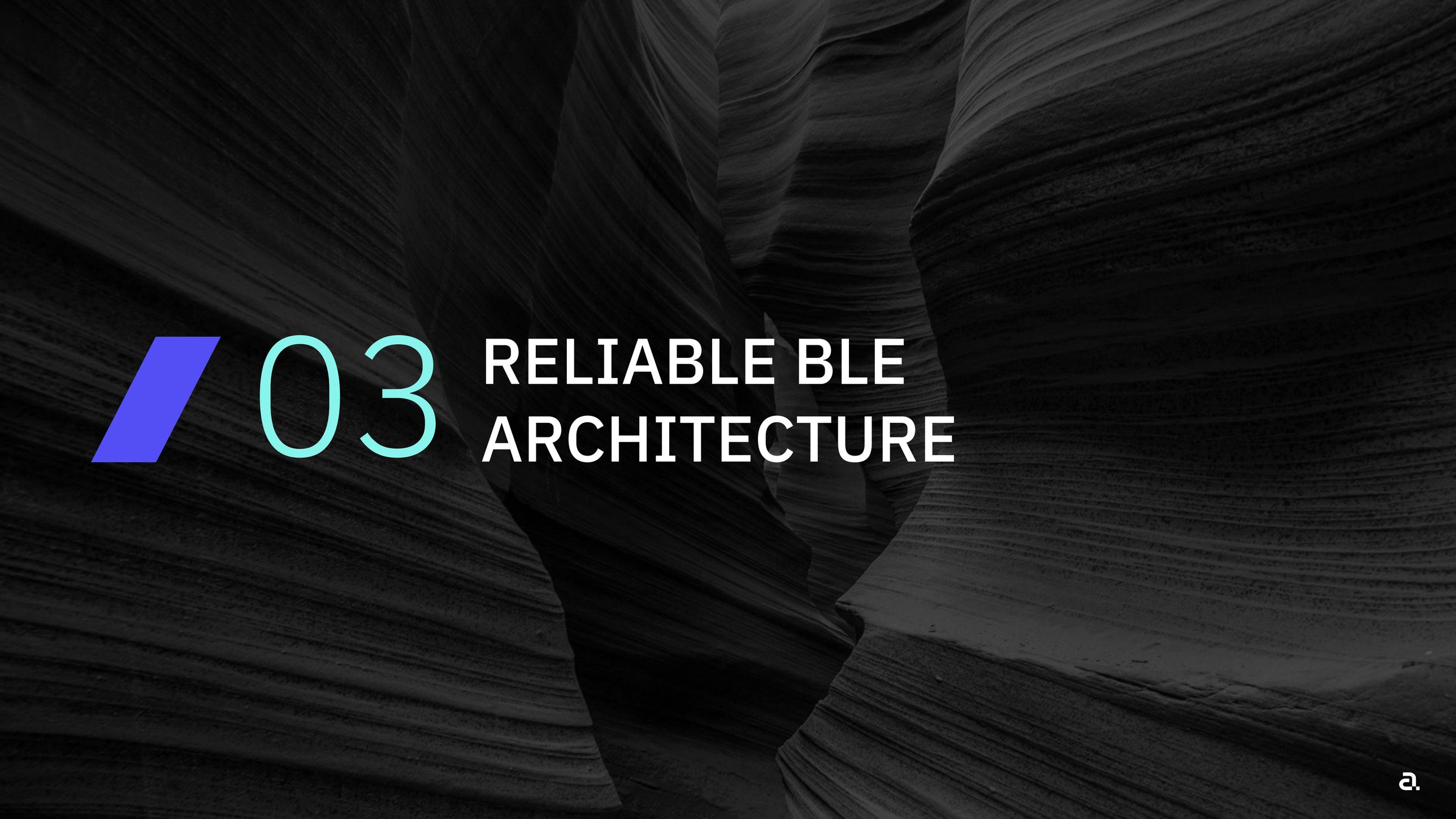
- Putting BLE logic inside UI controllers.

Problems:

- Tight coupling
- Difficult debugging
- Poor testability
- Hard to scale

Solution:

- Separate BLE communication from UI logic.



03 RELIABLE BLE ARCHITECTURE

RECOMMENDED ARCHITECTURE

```
BLE/
├── Protocol/
│   ├── BlePacket.swift
│   └── BleDeviceProtocol.swift
├── Transport/
│   └── BleTransport.swift
├── Manager/
│   ├── BleManager.swift
│   └── BleDeviceRegistry.swift
└── Devices/
    ├── DeviceA/ (Device A: constants, models, protocol)
    │   ├── DeviceAConstants.swift
    │   ├── DeviceAModels.swift
    │   └── DeviceAProtocol.swift
    └── DeviceB/(optional; Device B UUIDs only)
        └── DeviceBConstants.swift
```

Benefits:

- Loose coupling
- Easier debugging
- Testable
- Scales cleanly

04 RELIABILITY CHALLENGES

RANDOM DISCONNECTS

Apps must detect and recover from unexpected disconnects gracefully.

Common causes:

- Devices disconnect randomly
- Weak signals or interference
- Firmware bugs
- Background interruptions

RECONNECTION STRATEGY

Reliable apps automatically recover from disconnects.

Typical flow:

- Detect disconnect
- Wait a short delay
- Attempt reconnect
- Retry with limits

WRITE RELIABILITY ISSUES

BLE writes are unreliable without controlled message flow

BLE writes can fail due to:

- MTU size limits
- Packet collisions
- Device processing limits

COMMAND QUEUE PATTERN

Serializing commands prevents devices from being overwhelmed.

Flow:

Command queued

- Send command
- Wait for response
- Send next command

HANDLING FIRMWARE DIFFERENCES

Hardware evolves, so apps must support multiple firmware versions.

Challenges:

- Different firmware versions
- New features added
- Older devices still supported

Strategies:

- Firmware version check
- Feature flags
- Compatibility layers


05
TESTING &
DEBUGGING TOOLS

BLE DEBUGGING APPS

BLE debugging apps help isolate device problems quickly.

Tools:

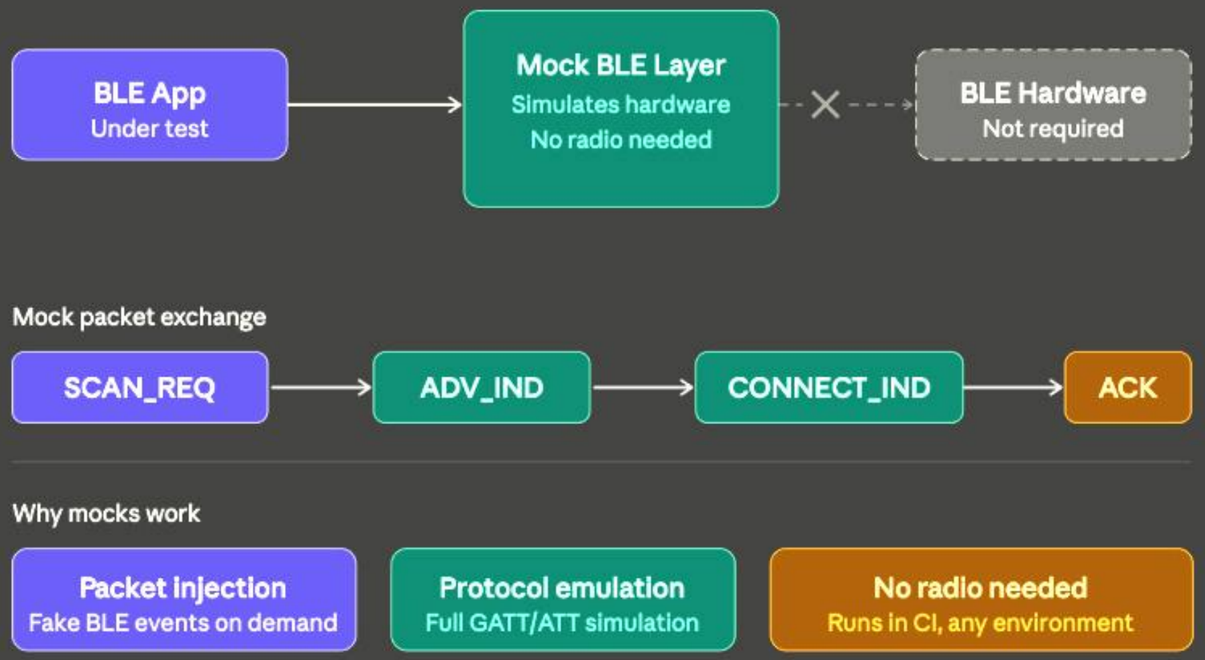
- LightBlue
- nRF Connect

MOCK BLE LAYER FOR TESTING

Mocks allow BLE apps to be tested without hardware.

Benefits:

- UI testing without real devices
- Automated testing
- Faster development cycles



DEBUGGING STRATEGIES

Combining logs, tools, and signal monitoring reveals most BLE issues.

Approaches:

- Monitor RSSI signal strength
- Track connection lifecycle
- Detect communication timeouts
- Inspect characteristic data


06
LIVE
DEMO

DEMO

Seeing the BLE lifecycle in action makes the concepts concrete.

07 PRODUCTION LESSONS

LESSONS FROM REAL BLE APPS

Design BLE apps assuming the connection will fail.

- Devices may behave unpredictably
- Firmware bugs are common
- Connections can drop unexpectedly
- Robust retry mechanisms are essential
- Strong logging and monitoring

Key Takeaways

Reliable BLE apps come from strong architecture and defensive design.

Reliable BLE apps require:

- Clear architecture
- State management
- Command queues
- Robust reconnection strategies
- Strong logging and monitoring

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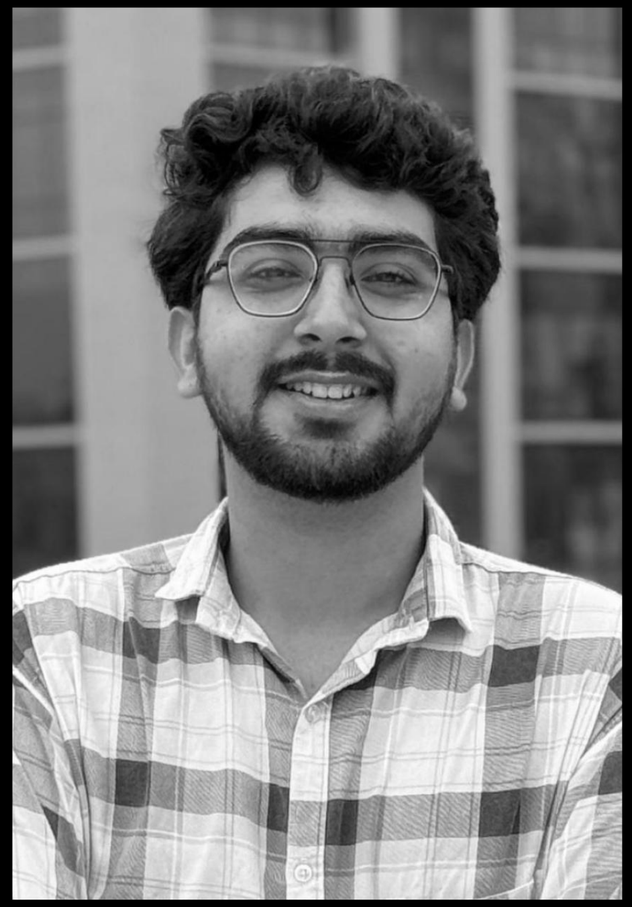
Thank You!



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